

Full length article

Pin-ended stainless steel equal-leg angle section members under compression and major-axis bending: Experimentation, FE modelling and design

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ABSTRACT

The behaviour and design of cylindrically-pinned stainless steel equal-leg angle section members under compression and compression combined with strong-axis bending are investigated herein. An experimental investigation, including material testing, initial geometric imperfection measurements and physical tests on hot-rolled stainless steel equal-leg angle section members is first presented. Numerical models are developed by means of shell finite element modelling formulated within ABAQUS and validated against experimental data. A numerical parametric study is then presented considering both hot-rolled and cold-formed stainless steel equal-leg angle section columns alongside beam-columns with a wide range of cross-section and member geometries. Finally, new design proposals for cylindrically-pinned stainless steel equal-leg angle section members under compression and compression plus major-axis bending are developed and verified against the results of physical experiments and numerical simulations. The proposed design rules are shown to offer substantially more accurate and consistent resistance predictions compared to existing codified design rules. The reliability of the new design provisions, with a recommended partial safety factor $\gamma_{M1} = 1.1$, is verified following the EN 1990 procedure.

1. Introduction

Angle section members are used in a range of structures, such as towers and trusses alongside the bracing systems in buildings and bridges. Even though early studies on angles date back a full century [1], their structural behaviour continues to pose challenges, with current design provisions in international standards known to have significant limitations [2–8]. New resistance functions for steel angle section members recently developed within the US [3,9,10] and European [4–7,11,12] design frameworks have led to substantial improvements in the consistency and accuracy of their load-carrying capacity predictions. However, little attention has been paid to pin-ended angle section columns under compression within the framework of Eurocode 3. The behaviour and design of angles under combined loading was investigated recently by Behzadi-Sofiani et al. [8], but this study was limited to combined compression and minor-axis bending. Angles are often loaded eccentrically (e.g. when connected through one leg), such that the line of action of loading does not coincide with the cross-section centroid (see Fig. 1), resulting in the member experiencing combined compression and bending about both principal axes. It is therefore crucial to study their behaviour when subjected to compression plus major-axis bending.

Behzadi-Sofiani et al. recently developed new design equations and procedures for fixed-ended steel and stainless steel angle section columns [4,5] and beams [6,7]. These proposals, along with an expression developed to predict the shift of the effective centroid due to local buckling [2], were then used to establish a new approach for designing pin-ended stainless steel equal-leg angle section members subjected to compression and combined compression and minor-axis bending [8]. The aim of the current paper is to develop new design proposals for pin-ended stainless steel equal-leg angle section members subjected to compression and combined compression and major-axis bending suitable for incorporation into EN 1993-1-4 [13]. The behaviour of such members differs from those studied previously since the shift of effective centroid does not have any impact when end-rotation about the minor axis is restrained. In addition, the interaction factors for angle section beam-columns buckling about the major axis are likely to differ from those buckling about the minor axis. A review of previous studies on equal-leg angle section members is presented in Section 2. An experimental investigation, involving material tests and tests on 7 austenitic stainless steel equal-leg angle section members under combined loading is described in Section 3. An extensive programme of numerical simulations, conducted using the commercial finite element

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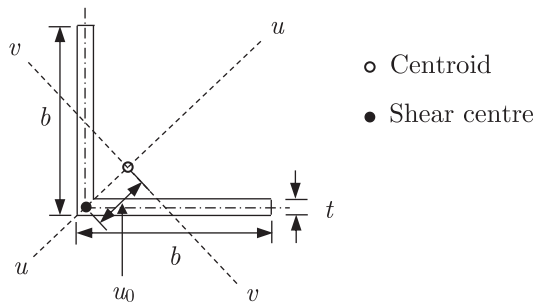


Fig. 1. Dimensions and principal axes of equal-leg angles.

(FE) software ABAQUS, is then presented in Section 4. The FE models are first validated against the results from the newly-conducted physical experiments and subsequently utilised to conduct a parametric study for producing additional data across a wide range of slenderness and torsional-flexural to minor-axis flexural elastic buckling load ratios; the observations are summarised in Section 5. Shortcomings in the current design provisions in EC3 are identified using the experimental and numerical results and are highlighted in Section 6. New proposals for the design of pin-ended stainless steel equal-leg angle section columns and beam-columns are subsequently presented and assessed against the established data. Statistical analyses are performed that confirm the accuracy and reliability of the proposed design method.

2. Review of previous research

Stainless steel equal-leg angle section columns have been the focus of a number of previous investigations, where their buckling behaviour has been studied under both fixed-ended [5,14–23] and pin-ended [8,19,20,23–29] support conditions, as well as bolted [30] and welded [31] end joints. The behaviour of stainless steel angle section members under uniform bending has also been investigated [7,22,32]. Previous tests on stainless steel angle section members are summarised in Table 1, where the manufacturing process, boundary and loading conditions as well as the number of tests are presented. As can be seen, the existing literature on pin-ended stainless steel equal-leg angle section columns is focused on columns pinned about the minor axis, the findings of which are described in [8]. To the best knowledge of the authors, there have been no experimental investigations reported on stainless steel equal-leg angle section members cylindrically pinned about the major axis, and subjected to compression or combined compression and bending. Therefore, in the current study, tests are carried out on hot-rolled stainless steel equal-leg angle section members under compression and combined loading, with these boundary conditions. Numerical models are then validated and are subsequently used to generate additional results to underpin the development of improved design rules for these members.

3. Experimental investigation

3.1. General

An experimental programme, involving material tensile coupon tests, initial geometric imperfection measurements and 7 pin-ended stainless steel angle section buckling tests on specimens subjected to compression and combined compression and major-axis bending, is presented in the current section. Residual stress measurements were not carried out, but representative residual stress distributions from the literature were utilised for the purposes of validation of the numerical models and in the parametric study, as described in Section 4. The tested cross-section profile was a hot-rolled $80 \times 80 \times 10$ equal-leg angle in Grade 1.4307 austenitic stainless steel; the chemical composition of the test material, as reported in the manufacturer's mill certificate, is presented in Table 2.

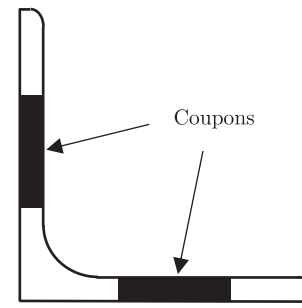


Fig. 2. Location of tensile coupons within the angle section.

3.2. Tensile coupon tests

Tensile coupon tests were conducted first to obtain the material properties of the studied stainless steel angle section. Two coupons were extracted from the middle portion of both legs in the longitudinal direction (see Fig. 2) following the dimensional requirements specified in EN ISO 6892 [33]. The tensile tests were conducted using an Instron 250 kN testing machine. The strains in the longitudinal direction were recorded using two linear electrical resistance strain gauges that were attached to the centre of the front and back faces of the coupons at mid-height. A clip gauge with a gauge length of 100 mm was placed within the parallel length; the extension was recorded and used to obtain the average strain over the gauge length. The tensile test set-up is shown in Fig. 3. The engineering stress–strain curves from the tensile coupon tests are presented in Fig. 4. The first stage of the stress–strain curves, up to the strain value of 10%, was obtained using the strain gauge readings, while the second stage of the curves until fracture was derived based on the clip gauge readings. The obtained average mechanical properties are presented in Table 3, where E is the Young's modulus, f_y and $f_{1.0}$ are the 0.2% and 1.0% proof stresses respectively, ϵ_u is the strain at the ultimate stress f_u , ϵ_f is the fracture strain measured over the 100 mm gauge length, while n , m_u and $m_{1.0}$ are the Ramberg–Osgood strain hardening exponents, with m_u and $m_{1.0}$ corresponding to the models described in [34–38], respectively. The measured 0.2% proof stress and the Young's modulus E are higher than the minimum specified values in [13], which are $f_y = 210 \text{ N/mm}^2$ and $E = 200,000 \text{ N/mm}^2$, respectively.

3.3. Initial geometric imperfection measurements

Initial geometric imperfections in the angle section specimens, including the bow imperfections about the minor and major axes, the amplitudes of which are denoted as $\delta_{0,v}$ and $\delta_{0,u}$, respectively, and the twist, of amplitude ϑ , along the member length, were measured prior to testing. The three modes of imperfection are presented in Fig. 5. The global bow and twist imperfections were determined using the same imperfection rig as employed in previous studies [5–8,39], where four LVDTs at the corner and the tips of the cross-section were utilised to measure the imperfections along the length of the members. The set-up and approach for measuring and calculating the initial imperfections were presented in detail in [5]. Typical measured longitudinal distributions of initial global and torsional geometric imperfections are presented in Fig. 6, where local imperfections of very low amplitude can also be observed. A half-sine wave was employed to represent the shape of the imperfections, with the amplitude being taken as the maximum measured deviation from straightness. A summary of the measured initial geometric imperfection amplitudes of the specimens relative to the member length L is provided in Table 4, which are lower than the assumed amplitude of $L/1000$ in the parametric study presented in Section 4.

Table 1
Summary of existing experiments on stainless steel equal-leg angle section members.

Reference	Boundary conditions	Loading condition	Axis of bending	Manufacturing process	No. of tests
Behzadi-Sofiani et al. [5]	Fixed-ended	Compression	–	Hot-rolling	5
Menezes et al. [14]				Hot-rolling	13
Liang et al. [15]				Hot-rolling	16
Sun et al. [16]				Hot-rolling	10
Sarquis et al. [17]				Hot-rolling	10
Sirqueira et al. [18]				Hot-rolling	18
Filipovic et al. [19]				Hot-rolling	6
Filipovic et al. [20]				Laser-welding	6
Zhang et al. [21]				Cold-forming	16
Zhang et al. [22]				Cold-forming	4
Dobric et al. [23]				Cold-forming	3
Zhang et al. [25]	Pin-ended	Compression and compression + bending	Minor axis	Hot-rolling	12
Zhang et al. [26]				Hot-rolling	12
Filipovic et al. [19]				Hot-rolling	20
Reynolds [24]				Laser-welding	33
Filipovic et al. [20]				Laser-welding	20
Dobric et al. [23]				Cold-forming	8
Behzadi-Sofiani et al. [8]				Hot-rolling	19
Behzadi-Sofiani et al. [7]		Bending	Minor axis	Hot-rolling	2
Behzadi-Sofiani et al. [7]			Major axis	Hot-rolling	5
Zhang et al. [22]			Geometric axis	Cold-forming	6
Theofanous et al. [32]			Geometric axis	Laser-welding	4
Total					248

Table 2
Chemical composition of tested cross-section from mill certificate.

Cross-section	C (%)	P (%)	S (%)	Si (%)	Mn (%)	Cr (%)	Ni (%)	Mo (%)
80 × 80 × 10	0.027	0.033	0.014	0.235	1.643	18.120	8.040	0.262

Table 3
Measured material properties obtained from tensile coupon tests.

Test	E (N/mm ²)	f_y (N/mm ²)	$f_{1.0}$ (N/mm ²)	f_u (N/mm ²)	ϵ_u (%)	ϵ_f (%)	n	m_u	$m_{1.0}$
Coupon 1	214 400	286	343	648	44.4	49.3	9.9	3.2	2.4
Coupon 2	205 000	292	348	654	43.7	48.5	11.9	3.3	2.6
Average	209 700	289	346	651	44.0	48.9	10.9	3.3	2.5

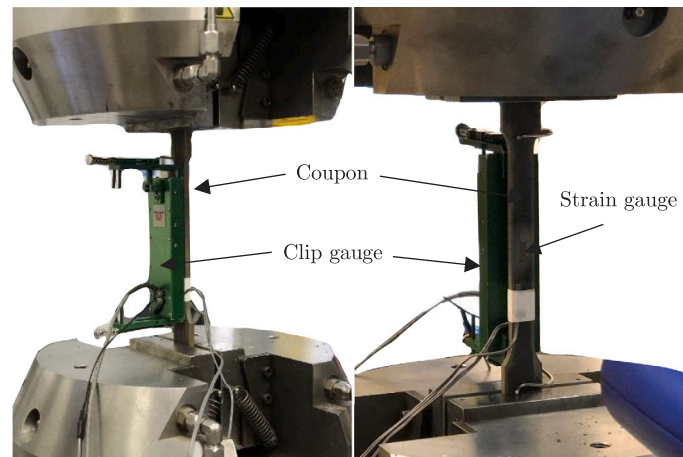


Fig. 3. Tensile coupon test set-up.

3.4. Member buckling tests

Tests on 7 hot-rolled Grade 1.4307 austenitic stainless steel equal-leg angle section members – 1 column and 6 beam-columns – were performed to study their buckling response and load-carrying capacity. The members were cylindrically pinned about the major axis i.e. free

to rotate about the major axis but fixed against rotation about the minor axis at the member ends. A member length L of 800 mm, for which torsional-flexural buckling was critical in compression, was selected for all tests. The initial loading eccentricities were varied to study the ultimate strength of stainless steel equal-leg angles subjected to a range of combinations of axial compression and bending.

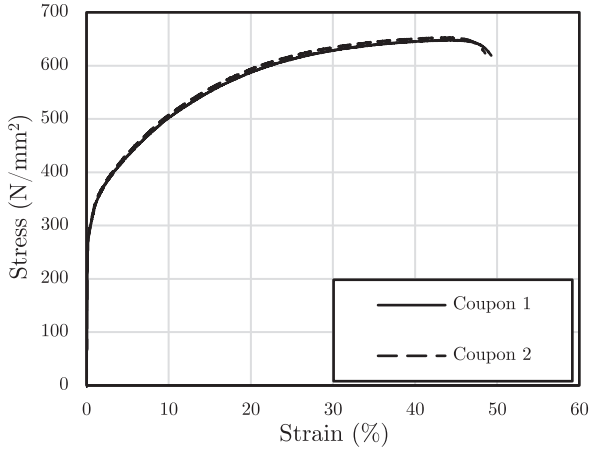


Fig. 4. Measured engineering stress–strain curves from tensile coupon tests.

Table 4

Summary of initial geometric imperfection amplitude measurements.

ID	$L/\delta_{0,v}$	$L/\delta_{0,u}$	θ/L (deg/m)
1	7273	2105	0.14
2	4705	5333	0.15
3	8000	1509	0.04
4	2667	2162	0.05
5	10 000	8889	0.01
6	2222	4444	0.08
7	3478	3077	0.04
Mean	5478	3931	0.07
CoV	0.54	0.66	0.78

The buckling tests were conducted using an Instron 2000 kN hydraulic testing machine under displacement control. Knife edges were used at both ends of the members to achieve the desired boundary conditions. End plates were welded to both ends of each specimen, which were inserted into special clamps connected to the knife edges. The specimens were then adjusted to achieve the required initial loading eccentricity, before the tightening of bolts on the clamp. One LVDT was used to measure the end-shortening of the specimens. Four string pots, attached to the corner and the tips of both legs of the angle section members, were employed to measure the twist and lateral deflection at mid-height. Four electrical resistance strain gauges were affixed to the specimens at mid-height to measure the longitudinal strains at the corner and the tips of the angles. Two inclinometers were used to measure rotations at the member ends. The configuration of the string pots and strain gauges is shown in Fig. 7 and the test set-up is presented in Fig. 8. The main test results, including the load, machine displacement and readings from the strain gauges, LVDT, string pots and inclinometers were recorded at 0.5 s intervals using the data acquisition equipment DATASCAN and logged using the DSLOG software.

A global imperfection amplitude of $L/1000$ is typically assumed in the development of column buckling design provisions [40–42]. Therefore, for the column buckling test, a total global imperfection amplitude ω_g about the major axis, defined as the sum of the initial bow imperfection $\delta_{0,u}$ and the initial loading eccentricity $e_{0,u}$, was set to $L/1000$ by adjusting the initial loading eccentricity. This was achieved by pre-loading the specimen up to 20% of the predicted ultimate load and back calculating the total global imperfection ω_g , thus:

$$\omega_g = \frac{EI(\epsilon_{\max} - \epsilon_{\min})}{HN} - \delta, \quad (1)$$

where I is the second moment of area about the axis of buckling (i.e. the major axis), $\epsilon_{\max}$ and $\epsilon_{\min}$ are the average strains recorded by the strain gauges on the concave and convex sides of the specimens,

respectively, N is the applied axial load, H is the distance between the strain gauges and δ is the measured lateral deflection at the mid-height. The initial loading eccentricity was then adjusted and the procedure repeated until ω_g reached a value close to $L/1000$.

For the beam–column test specimens, ω_g was back calculated after the experiments using a similar procedure. Note that for the beam–columns, ω_g denotes the total loading eccentricity, with the initial loading eccentricity $e_{0,u}$ dwarfing the initial bow imperfection $\delta_{0,u}$. A typical plot of load N against ω_g , shown for Specimen 4, is presented in Fig. 9, where $\omega_g = 23.0$ mm was determined. The geometry and notation of the test specimens are presented in Fig. 10, while a summary of the measured geometric properties of the test specimens and the main results from the column and beam–column buckling tests are presented in Table 5, noting that $N_{cr,TF} = 1800$ kN (calculated based on the measured geometry and Young's modulus). Graphs of the test axial load versus end-shortening and mid-height lateral deflection responses of all specimens are plotted in Fig. 11. Specimen 1 ($w_g = 0.5$ mm) failed by torsional-flexural buckling, with the direction of flexure dictated by the direction of the initial geometric imperfection. For the remainder of the test specimens, which were subjected to combined axial load and major-axis bending, failure was dominated by major-axis flexure, with little or no evidence of twist (see Fig. 12). In Eurocode 3, the combined effect of axial compression and bending moment is checked through the N – M interaction curve, the two ends of which are anchored to the axial compression N and bending moment M resistances of the member. The shape of the interaction curve depends on the normalised slenderness, as well as the cross-section type and material grade; this is further explained in Section 6. Comparisons of the test data with the N – M interaction curve for I-sections (since there is no guidance for angle section members) set out in prEN 1993-1-4 [43], are presented in Fig. 13, where the ultimate load N_u and moment M_u are normalised against the compression $N_{b,EC3}$ and bending moment $M_{b,EC3}$ resistances. The test results generally follow the trend of the N – M interaction curve and lie consistently on the safe side.

4. Numerical modelling

The commercial finite element (FE) software ABAQUS was used to create numerical models to simulate the mechanical behaviour of stainless steel equal-leg angle section members cylindrically pinned about the major axis and subjected to compression and combined compression and bending. The FE models are first validated in Section 4.2 through comparisons against the newly-generated experimental data presented herein. In Section 4.3, a parametric study is presented to investigate the behaviour of both hot-rolled and cold-formed stainless steel equal-leg angle section members cylindrically pinned about the major axis and subjected to compression and combined loading; a range of cross-section geometries, member lengths and load combinations are studied.

4.1. General modelling assumptions

Similar modelling assumptions to those used in [4–8] were adopted, which are briefly described herein. The 4-noded shell element, S4R [44], with reduced integration, 6 degrees of freedom (3 translational and 3 rotational) at each node and hourglass control, was employed to model the angle section members; this element type has been successfully utilised in similar previous studies [45–48]. A mesh size of approximately 5 mm, typically resulting in at least 10 elements across each leg of the angles, was utilised.

Both hot-rolled and cold-formed stainless steel angles were modelled. For the hot rolled angles, the root fillet was modelled by introducing thicker elements at the junction between the angle section legs and the thickness was offset to avoid overlapping of the two legs, which were connected using BEAM-type multi-point constraints (MPC), as shown in Fig. 14(a). The rounded corner of the cold-formed angles

Table 5
Summary of measurements of geometric properties, ω_g and ultimate loads of test specimens.

Cross-section	ID	b (mm)	t (mm)	r_1 (mm)	r_2 (mm)	L (mm)	$L_{cr,u}$ (mm)	$\bar{\lambda}_{TF}$	ω_g (mm)	N_u (kN)
$80 \times 80 \times 10$	1	80.26	9.81	10	5	800	950	0.49	0.5	424.7
	2	80.29	9.81	10	5	798	948	0.49	7.2	357.3
	3	80.14	9.81	10	5	798	948	0.49	9.3	345.3
	4	80.25	9.80	10	5	800	950	0.49	23.0	261.6
	5	80.26	9.79	10	5	800	950	0.49	56.2	170.7
	6	80.30	9.77	10	5	801	951	0.49	111.0	103.3
	7	80.20	9.78	10	5	798	948	0.49	221.0	57.8

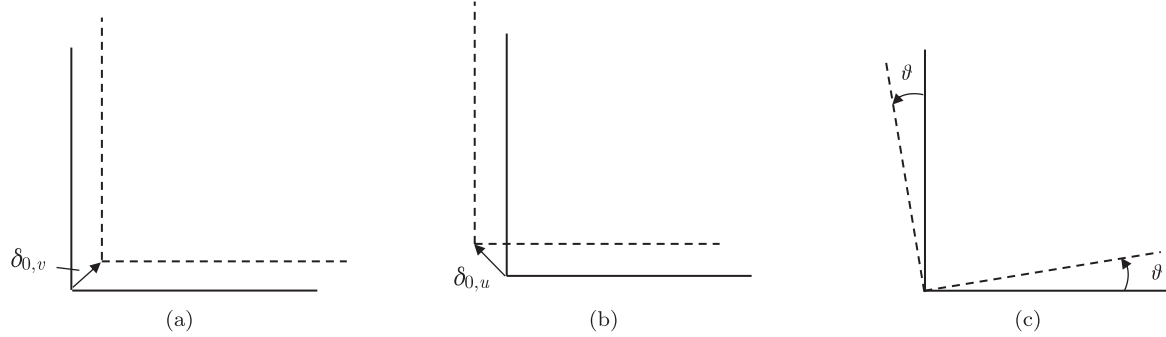


Fig. 5. Illustrations of amplitudes of imperfection shapes for (a) minor-axis flexural, (b) major-axis flexural and (c) torsional modes.

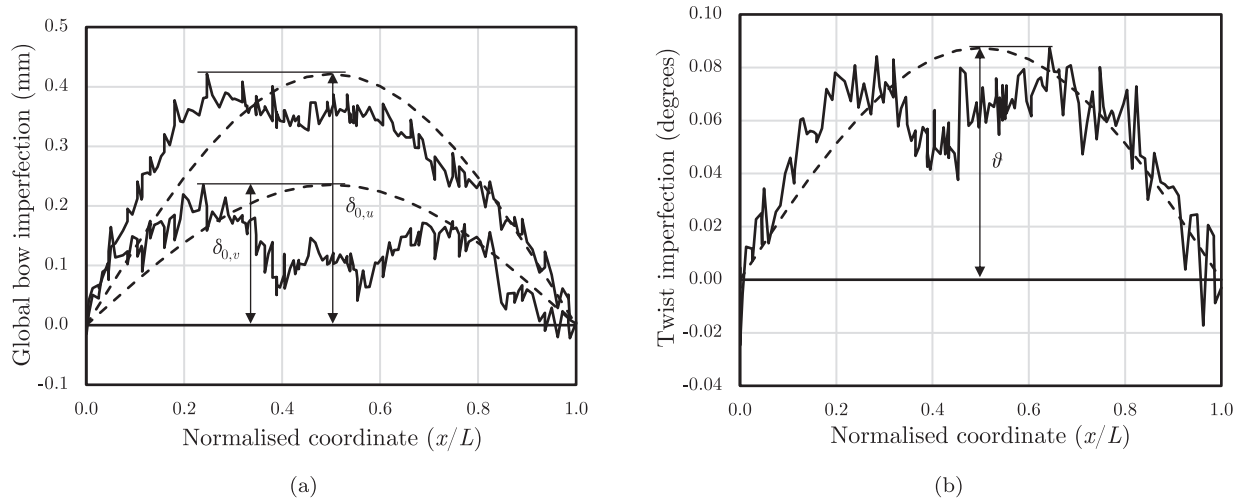


Fig. 6. Typical measured initial (a) global bow and (b) twist geometric imperfections versus normalised position along the member length (x/L), with dashed lines marking representative half-sine waves.

was represented by employing a finer mesh in the curved corner region than in the flat regions, as shown in Fig. 14(b). To model the load applied to the members, a longitudinal displacement was imposed at one end through a reference point that was free to move longitudinally. The geometrically and materially nonlinear analyses were executed using the Riks arc-length method [49]. To validate the FE models, the experiments reported in the current study were simulated. The measured geometric properties and stress–strain curves were employed in these analyses. Throughout the parametric study, the two-stage Ramberg–Osgood material model [34–37], with typical values [50] for the key mechanical properties, was utilised. For the cold-formed angles, an enhanced yield strength in the corners, caused by work hardening during forming [51,52], was also considered. A Poisson's ratio of 0.3 in the elastic range was assumed in all cases. The material properties were input into ABAQUS in the form of true stress σ_{true} and true plastic strain ϵ_{true} , converted from the measured engineering stress σ and strain ϵ using the well-known relationships:

$$\sigma_{true} = \sigma(1 + \epsilon), \quad (2)$$

$$\epsilon_{true} = \ln(1 + \epsilon) - \frac{\sigma_{true}}{E}. \quad (3)$$

For the hot-rolled stainless steel angles, a bilinear residual stress distribution with a peak value of 70 MPa, constant with thickness and leg width, was employed [4–7], as illustrated in Fig. 15, where the negative sign indicates compressive stress and the positive sign denotes tensile stress. The residual stresses were input into ABAQUS using the *INITIAL CONDITIONS command. For the cold-formed stainless steel angles, the dominant bending residual stresses were assumed to be inherently captured in the material stress–strain curves [53,54], which had a lower value for the strain hardening exponent n than for hot-rolled material, and were therefore not explicitly defined. For the validation study, the measured initial minor-axis global bow and twist imperfection amplitudes, as presented in Table 4, were applied; about the major axis, the measured loading eccentricity values given in Table 5, which included the major-axis global bow imperfections, were employed. However, throughout the parametric study, a half-sine wave function over the member length with an amplitude of $L/1000$ at mid-span was adopted about both principal axes for the initial bow

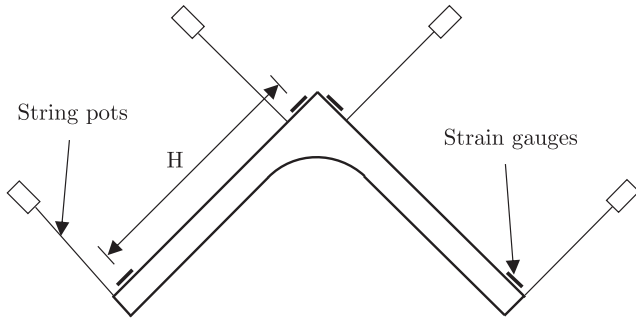


Fig. 7. Configuration of strain gauges and string pots at the mid-height of the column specimens.

Table 6

Summary of comparisons of FE model ultimate loads $N_{u,FE}$ with those obtained experimentally $N_{u,Test}$.

Source	No. of tests	$\frac{N_{u,FE}}{N_{u,Test}}$			
		Mean	CoV	Min	Max
Present study	7	0.96	0.02	0.94	1.00

imperfections. The direction of the major-axis bow imperfection was chosen to provide the lowest resistance in the parametric study, which was in the same direction as bending (i.e. in the opposite direction to the initial eccentricity) for the beam-columns. A similar shape was also adopted for the initial twist with an amplitude of $\theta = \tan^{-1}(L/1000b)$ at mid-span. Boundary conditions were applied to the supports through kinematic coupling constraints to link all the degrees of freedom (translational and rotational) of the member end-nodes to two reference points. Cylindrically pin-ended boundary conditions were created by restraining the necessary degrees of freedom at the reference points, where longitudinal displacement (i.e. end-shortening) at the loaded end and rotation about the axis of bending (i.e. the major axis) at both ends were free and the remaining degrees of freedom were fixed; warping was also prevented at both ends.

4.2. Validation

The FE models were validated against the 7 tests from the present study; a summary of the comparisons between the ultimate resistances determined from the FE models and those obtained experimentally is presented in Table 6 and Fig. 16, where R_u signifies the test/FE ultimate capacity under combined loading, defined thus:

$$R_u = \sqrt{N_u^2 + M_u^2}, \quad (4)$$

and $R_{b,Rd}$ is the resistance under combined loading, given by:

$$R_{b,Rd} = \sqrt{N_{b,Rd}^2 + M_{b,Rd}^2} \quad (5)$$

with $R_{b,EC3}$ and $R_{b,prop}$ referring to the EC3 [43] and proposed resistances, respectively. The angle θ is illustrated in Fig. 17 and is calculated as:

$$\theta = \tan^{-1} \left(\frac{M_u/M_{b,Rd}}{N_u/N_{b,Rd}} \right) = \tan^{-1} \left(\frac{N_{b,Rd}e_0}{M_{b,Rd}} \right). \quad (6)$$

Overall, there is good agreement between the test and FE ultimate loads, with a mean $R_{u,FE}/R_{u,Test}$ ratio of 0.96 and a coefficient of variation (CoV) of 0.02. In addition to the FE model validation presented herein, the developed FE model has also been demonstrated to be suitable for simulating the structural response of equal-leg angle section

Table 7

Summary of ranges of key varied parameters in FE parametric study.

b/t	L/b	$N_{cr,TF}/N_{cr,F,v}$	$\bar{\lambda}_{TF}$ or $\bar{\lambda}_v$
8–30	1–60	0.05–1.10	0.2–3.0

members with different material characteristics under various loading conditions in several recent studies [4–8].

A comparison between typical test and FE load–deformation curves, shown for Specimens 5 and 7 tested in the current study, are presented in Fig. 18, illustrating good agreement between the experimentally-obtained and numerically-simulated results. Note that the earlier deviation from linearity of the FE curves than the test curves in Fig. 18 is believed to be due to residual stress magnitudes in the FE models being higher than those in the test specimens. Additionally, the deformed shapes of the test specimens were compared to those from the numerical models, as shown in Fig. 19 for Specimen 7 tested in the present study, and found to be in agreement. The described comparisons demonstrate the ability of the developed FE models to predict the physical behaviour observed in experiments on pin-ended stainless steel angle section columns and beam-columns, and confirm their suitability for performing a parametric study.

4.3. Parametric study

Following validation of the numerical models, a parametric study was conducted, as presented currently. The FE models were used to generate additional data for hot-rolled and cold-formed pin-ended angle section columns and beam-columns in the three main families of stainless steel (i.e. austenitic, duplex and ferritic), considering a wide range of cross-section geometries, member lengths and load combinations. A summary of the material properties adopted in the parametric study, taken from [50], is presented in [5]. Leg widths b of 50 mm and 100 mm were analysed, while the thicknesses t and member lengths L were varied to evaluate a spectrum of normalised torsional-flexural $\bar{\lambda}_{TF} = \sqrt{Af_y/N_{cr,TF}}$ or minor-axis flexural $\bar{\lambda}_v = \sqrt{Af_y/N_{cr,F,v}}$ slendernesses, in which A is the cross-sectional area, f_y is the yield strength, $N_{cr,TF}$ is the elastic torsional-flexural buckling load and $N_{cr,F,v}$ is the elastic minor-axis flexural buckling load, and elastic buckling load $N_{cr,TF}/N_{cr,F,v}$ ratios. The b/t ratios ranged between 8 and 30, while the L/b ratios ranged between 1 and 60 (see Table 7). In addition, the initial eccentricities were varied to study a wide range of load combinations. In total, 2100 FE results (350 for each of the 6 production type and stainless steel family combinations) were generated. The results are presented in Fig. 20, in which the normalised axial load $N_u/N_{b,EC3}$ (where N_u is the ultimate load obtained from the tests or FE models and $N_{b,EC3}$ is the axial compression resistance of the members predicted by EC3) is plotted against the corresponding normalised bending moment $M_u/M_{b,EC3}$ (where $M_u = N_u e_0$ is the ultimate bending moment obtained from the tests or FE models and $M_{b,EC3}$ is the bending resistance of the members predicted by EC3). For reference, the interaction curve from prEN 1993-1-4 [43] for austenitic stainless steel I-section members (since there is no guidance for angle section members) with $\bar{\lambda}_v = 3.0$ is also shown in the figures.

5. Analysis and discussion of results

As can be seen in Fig. 20, some of the data points on the vertical axis (i.e. members under axial compression) are on the unsafe side. This is contrary to the observations made in the previous studies [4,5], where the EC3 predicted buckling loads were conservative for fixed-ended equal-leg angle section columns. It was shown in [8] that when bending about the minor axis is allowed, the buckling resistance of equal-leg angle section columns is reduced due to the shift of the effective centroid. However, this is not the case for angles cylindrically

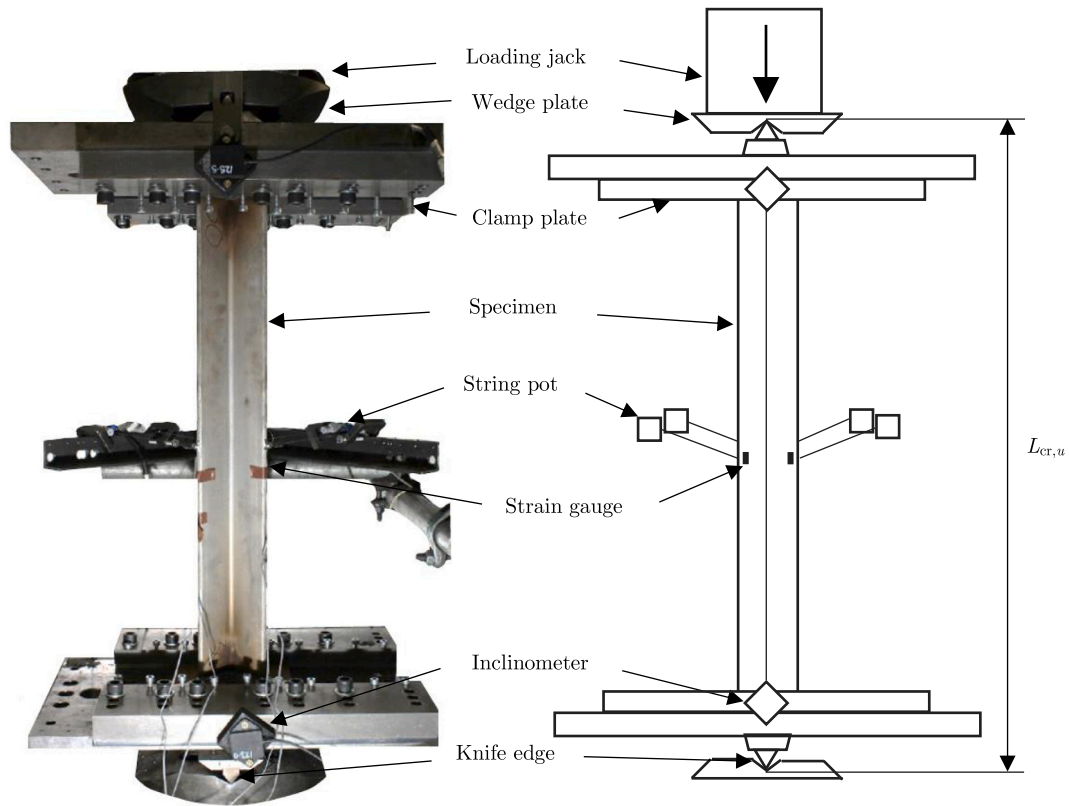


Fig. 8. Photographic and schematic representation of column and beam-column buckling test set-up.

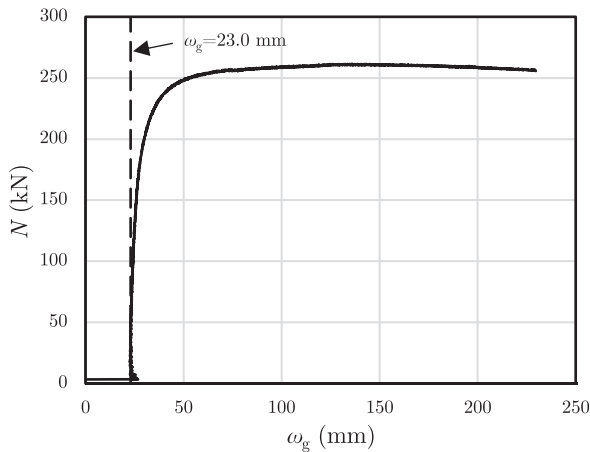


Fig. 9. Typical plot of load N against ω_g for Specimen 4, where $\omega_g = 23.0$ mm.

pinned about the major axis, since the shift of the effective centroid only occurs about the minor axis, owing to their symmetry about the major axis. The obtained results are analysed in the present section to gain insight into the observed behaviour.

The variation of the elastic buckling load N_{cr} with member length L (on a logarithmic scale), considering the different buckling modes, is presented in Fig. 21 for a typical equal-leg angle section column cylindrically pinned about the major axis. It can be seen that the minor- and major-axis flexural buckling loads are very similar ($\bar{\lambda}_v \approx \bar{\lambda}_u$), which is not the case for previously studied boundary conditions, suggesting that the two global flexural buckling modes could be triggered simultaneously. In addition, a study into the elastic post-buckling stability of equal-leg angle section columns, where both ends are fixed about the minor axis, and elastically restrained against rotation about the

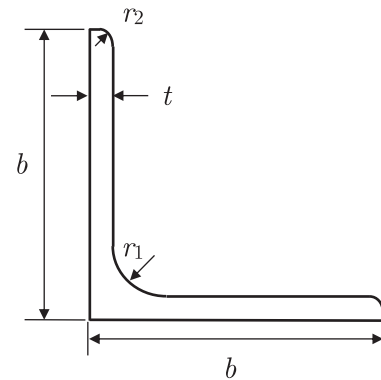


Fig. 10. Geometry and notation for equal-leg angle sections.

major axis by a rotational spring, the stiffness k of which varies between zero (i.e. simply supported) and fixed conditions, was conducted. The relationship between normalised load N/N_{cr} and normalised axial shortening δ/δ_{cr} , where δ_{cr} is the axial shortening at the elastic buckling load N_{cr} , is presented in Fig. 22 for an angle section column with slenderness $\bar{\lambda}_{TF} = 2.5$ but varying rotational stiffness about the major axis. The varying support stiffness about the major axis is also reflected in the $N_{cr,F,u}/N_{cr,F,v}$ ratios, which increase with increasing rotational stiffness. The dramatic variation in elastic post-buckling behaviour is clear, where the curves associated with columns with lower $N_{cr,F,u}/N_{cr,F,v}$ ratios (<2.0) display increasingly negative post-buckling stiffness, which signifies more severe unstable post-buckling.

To investigate the influence of the $N_{cr,F,u}/N_{cr,F,v}$ ratio on the buckling resistance of cylindrically-pinned angle section columns further, the relationship between the ratio $(\chi_{TF} - \chi_F)/(\chi_T - \chi_F)$, in which

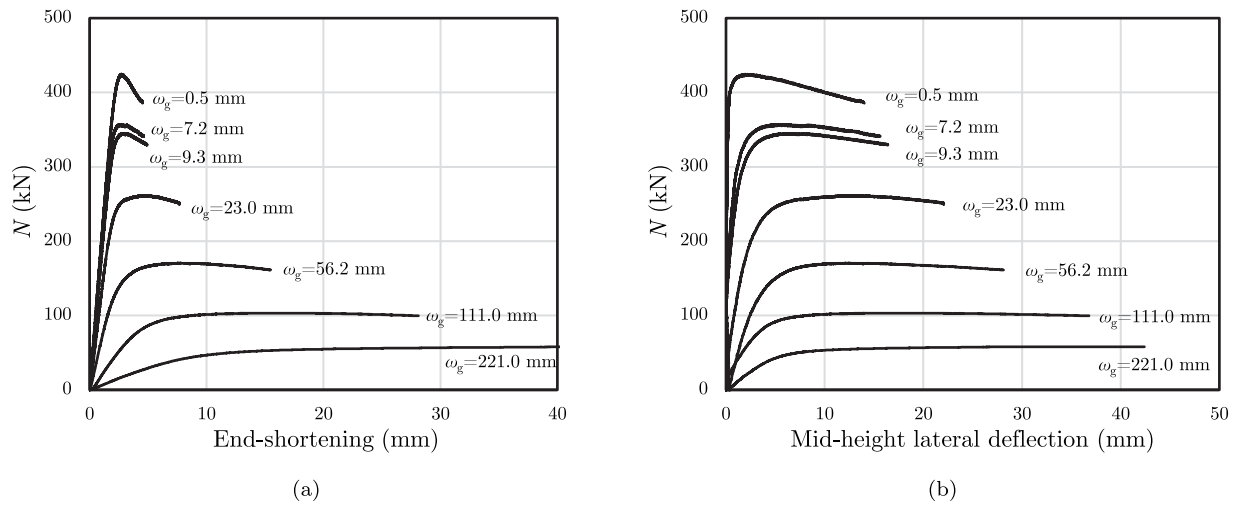


Fig. 11. Axial load N versus (a) end-shortening and (b) mid-height major-axis lateral deflection curves for the test specimens, the eccentricities ω_g of which are indicated on the figure.

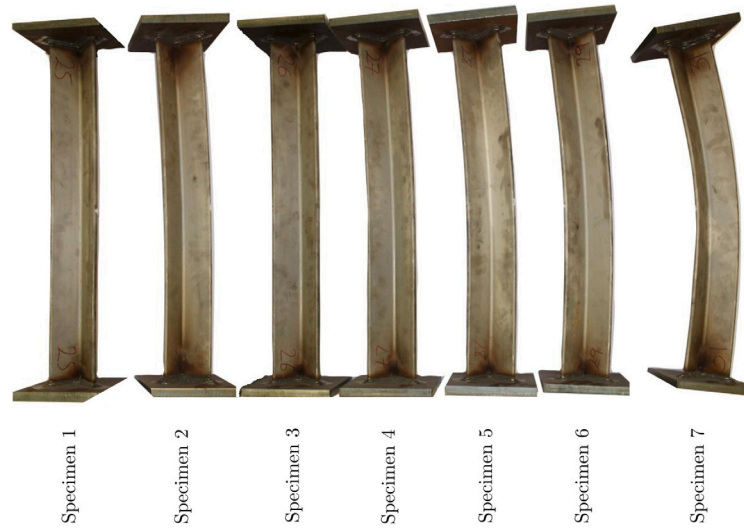


Fig. 12. Deformed column and beam-column test specimens.

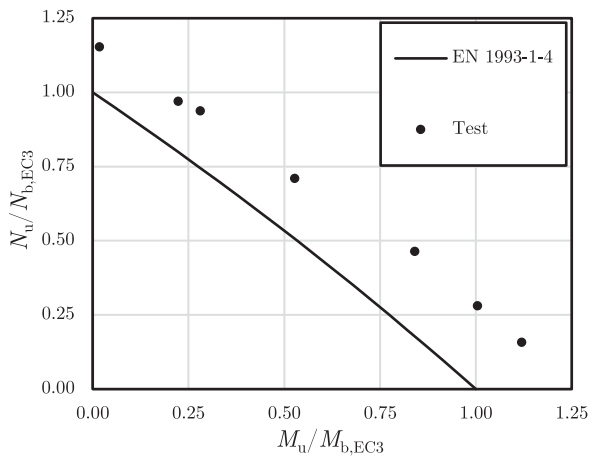


Fig. 13. Comparison of test data with prEN 1993-1-4 [43] N - M interaction curve.

$\chi_{TF} = N_{u,FE} / A f_y$ is the torsional-flexural reduction factor determined from the FE simulations, χ_T is the torsional reduction factor based on the local buckling curve given in EN 1993-1-5 [55], where ρ is replaced with χ_T and $\bar{\lambda}_{TF}$ is used instead of $\bar{\lambda}_p$, χ_F is the flexural reduction factor based on the flexural buckling curve given in EN 1993-1-1 [13], and the $N_{cr,TF} / N_{cr,F,v}$ values corresponding to the cases presented in Fig. 22, is plotted in Fig. 23(a). Note that the ratio $(\chi_{TF} - \chi_F) / (\chi_T - \chi_F)$ signifies where the torsional-flexural buckling reduction factor lies between the flexural and local buckling curves [4]. As can be seen, the minimum value of $(\chi_{TF} - \chi_F) / (\chi_T - \chi_F)$ reduces with decreasing rotational spring stiffness. This means that the level of interaction between torsional-major-axis-flexural and minor-axis-flexural buckling, which was observed in fixed-ended angle section columns [4,5], is even more significant when the rotational restraint about the major axis is reduced. This relationship may also be shown through the $N_{cr,F,u} / N_{cr,F,v}$ ratio, where a newly defined term β (as described in Section 7.2), which accounts for the reduction in buckling resistance due to the aforementioned interaction, varies with $N_{cr,F,u} / N_{cr,F,v}$, with a maximum magnitude at $N_{cr,F,u} / N_{cr,F,v} = 1.0$ (i.e. where the highest level of interaction is observed among the different boundary conditions) and a minimum value at $N_{cr,F,u} / N_{cr,F,v} = 4.0$. The variation of $(\chi_{TF} - \chi_F) / (\chi_T - \chi_F)$ is also plotted against $N_{cr,F,u} / N_{cr,F,v}$

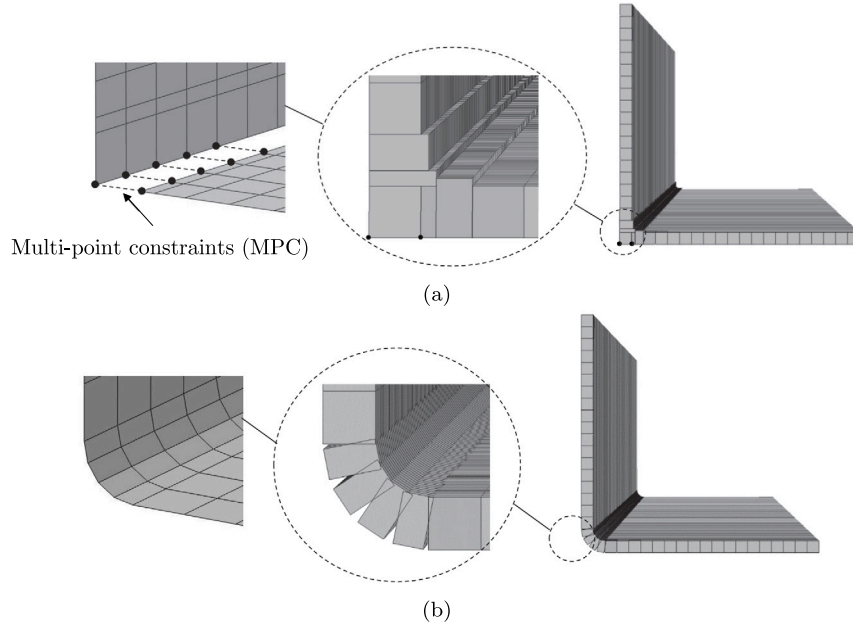


Fig. 14. FE representation of corner region in (a) hot-rolled and (b) cold-formed angle section members. Further details are provided in [5].

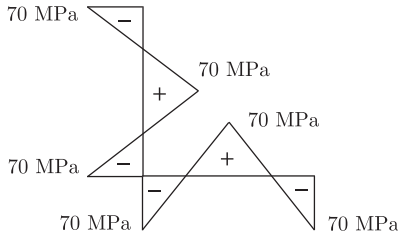


Fig. 15. Residual stress distribution employed for hot-rolled stainless steel equal-leg angles in the present FE models.

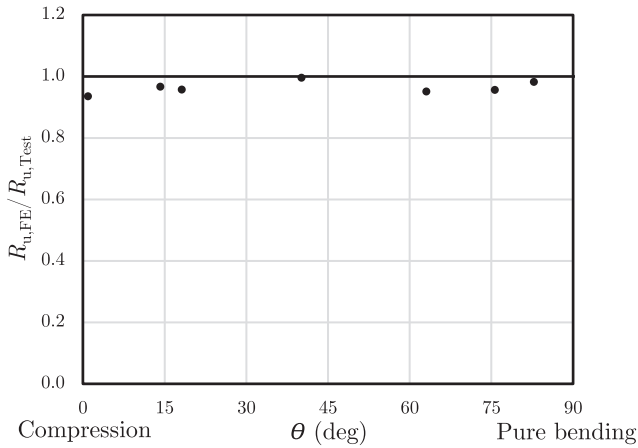


Fig. 16. Variation of ratios of FE model ultimate resistances $R_{u,FE}$ to experimental ultimate resistances $R_{u,Test}$ with angle θ .

in Fig. 23(b) for different $N_{cr,TF}/N_{cr,F,v}$ ratios, which shows reducing $(\chi_{TF} - \chi_F) / (\chi_T - \chi_F)$ values with reducing $N_{cr,F,u}/N_{cr,F,v}$ ratios for all $N_{cr,TF}/N_{cr,F,v}$ values. The aforementioned observations dictate the necessity for a new definition for β compared to that proposed in [5] to capture the influence of different boundary conditions on the ratio of global flexural buckling loads hence the severity of interactive buckling.

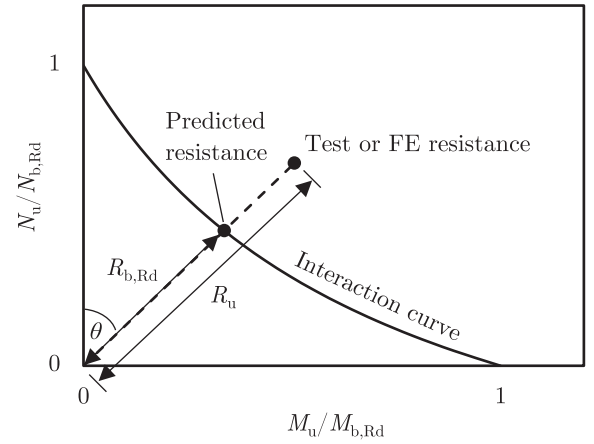


Fig. 17. Definition of parameters R_u , $R_{b,Rd}$ and θ .

6. Design to Eurocode 3

Angle section compression members cylindrically pinned about the major axis are not affected by the shift of the effective centroid (due to local buckling in Class 4 cross-sections), because the shift only creates an eccentricity relative to the fixed (minor) axis. Hence, no additional bending is induced and the EC3 member buckling resistance $N_{b,Rd}$ is simply given by [43]:

$$N_{b,Rd} = \frac{\chi A f_y}{\gamma_{M1}}, \quad (7)$$

where A is the cross-sectional area, which is replaced by the effective area A_{eff} for Class 4 cross-sections [55], χ is the buckling reduction factor from the relevant member buckling curve and γ_{M1} is the partial factor for member buckling, taken as $\gamma_{M1} = 1.1$ for stainless steel. According to prEN 1993-1-4 [43], members subjected to combined loading with bending about the major $u-u$ axis and axial compression should satisfy the following criteria:

$$\frac{N_{Ed}}{\chi_u N_{Rk}} + k_{uu} \frac{M_{u,Ed} + \Delta M_{u,Ed}}{\chi_{LT} M_{u,Rk}} \leq 1.0. \quad (8)$$

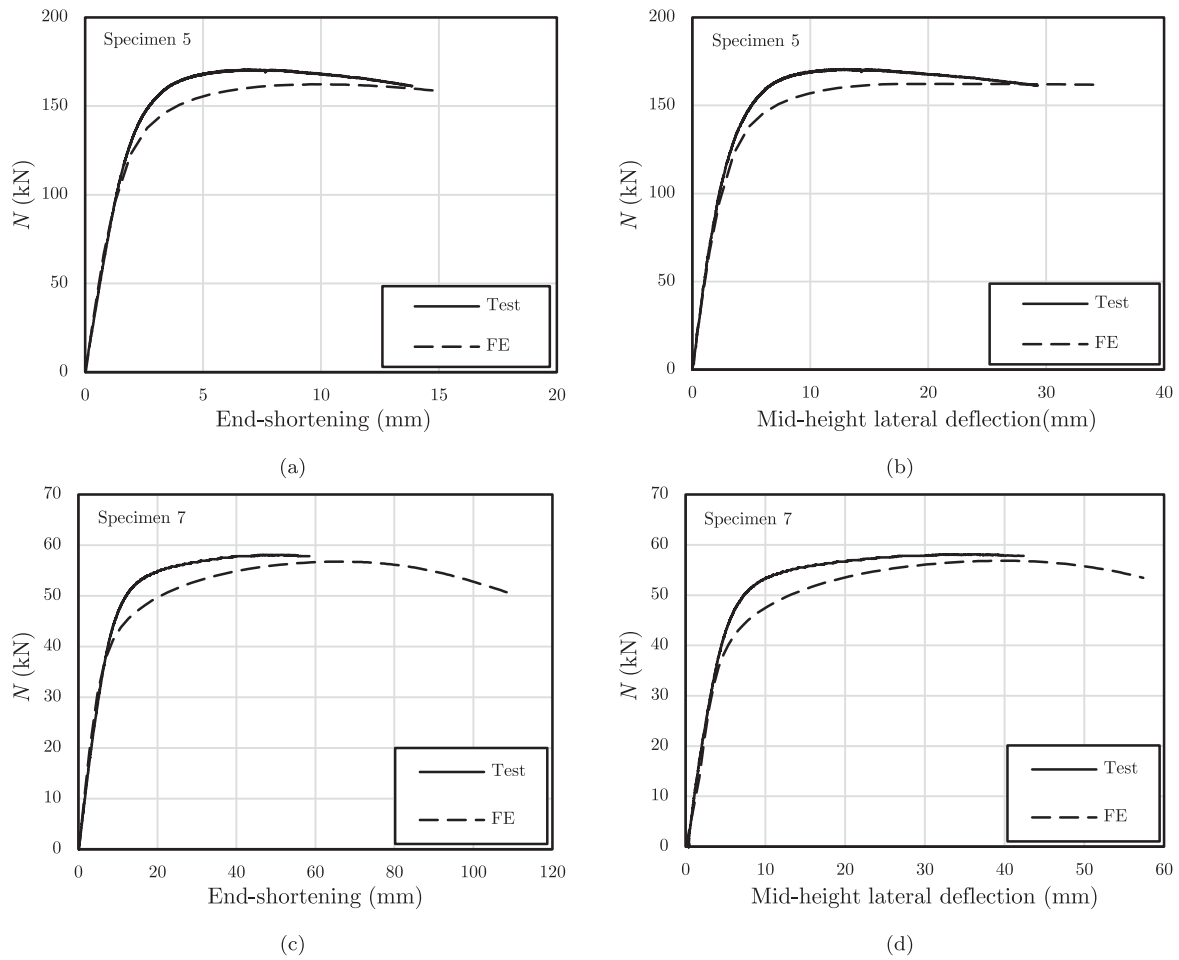


Fig. 18. Comparison of test and FE load N versus (a, c) end-shortening and (b, d) mid-height major-axis lateral deflection curves for Specimens 5 and 7 examined in the present study.



Fig. 19. Comparison of typical member deformed shape between (a) test and (b) FE model for Specimen 7 tested in the present study.

and

$$\frac{N_{Ed}}{\chi_v N_{Rk}} + k_{vu} \frac{M_{u,Ed} + \Delta M_{u,Ed}}{\chi_{LT} M_{u,Rk}} \leq 1.0. \quad (9)$$

Here, N_{Ed} and $M_{u,Ed}$ are the design values of the compression force and maximum bending moment about the major $u-u$ axis along the member

length, respectively, N_{Rk} and $M_{u,Rk}$ are the characteristic values of the cross-sectional resistance to compressive axial force and bending moment about the major $u-u$ axis, respectively, and χ_{LT} is the lateral-torsional buckling reduction factor. The quantity $\Delta M_{u,Ed}$ is the bending moment that arises due to the shift of the effective centroid about the major $u-u$ axis, $e_{N,u}$, for Class 4 cross-sections, which is zero for equal-leg angle section members, owing to their symmetry about the

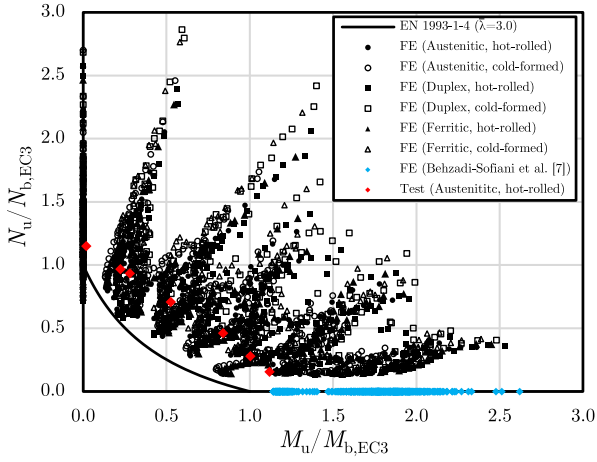


Fig. 20. Comparison of test and FE results for stainless steel equal-leg angle section columns and beam-columns cylindrically pinned about the major axis against EC3 interaction curve.

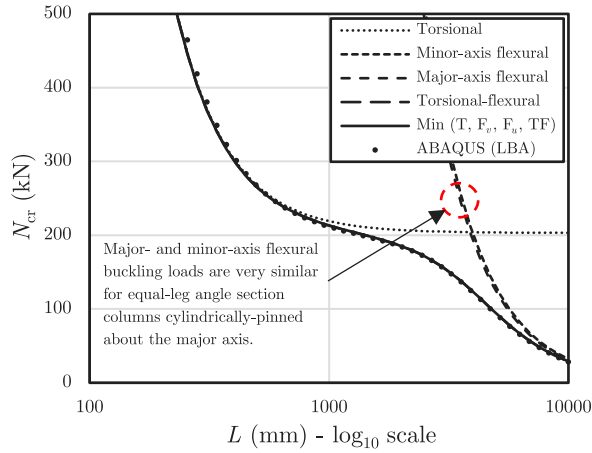


Fig. 21. Elastic buckling loads N_{cr} for the different modes plotted against member length L for an equal-leg angle section ($100 \times 100 \times 5$) column cylindrically pinned about the major axis.

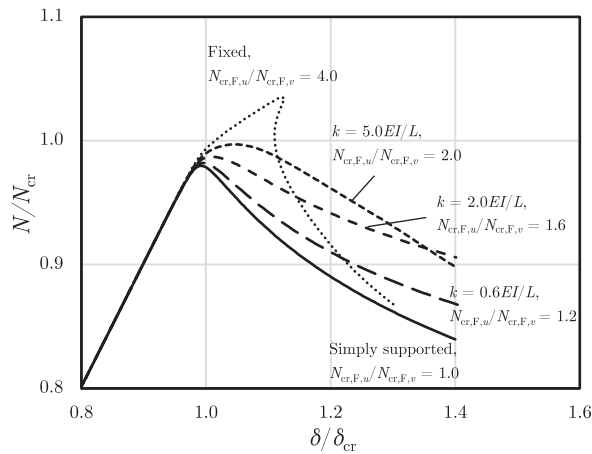


Fig. 22. Relationship between normalised load N/N_{cr} and axial shortening δ normalised by the axial shortening at the elastic buckling load δ_{cr} for an angle section column with slenderness $\lambda_{TF} = 2.5$ but varying rotational stiffness about the major axis at the member ends.

major axis. The terms χ_u and χ_v are the major- and minor-axis flexural buckling reduction factor, respectively, where χ_v should be replaced by χ_{TF} for members where torsional-flexural buckling is critical (i.e. when $N_{cr,TF} \leq N_{cr,F,v}$), while k_{uu} and k_{vu} are the interaction factors defined, for the case of uniform bending, thus:

$$k_{uu} = \begin{cases} 1 + 2.5 (\bar{\lambda}_u - 0.35) n_u & \text{for } \bar{\lambda}_u < 1.0 \\ 1 + 1.625 n_u & \text{for } \bar{\lambda}_u \geq 1.0 \end{cases} \quad \text{for austenitic stainless steel,} \quad (10)$$

$$k_{uu} = \begin{cases} 1 + 2.0 (\bar{\lambda}_u - 0.30) n_u & \text{for } \bar{\lambda}_u < 1.3 \\ 1 + 2.0 n_u & \text{for } \bar{\lambda}_u \geq 1.3 \end{cases} \quad \text{for duplex stainless steel,} \quad (11)$$

$$k_{uu} = \begin{cases} 1 + 1.6 (\bar{\lambda}_u - 0.35) n_u & \text{for } \bar{\lambda}_u < 1.3 \\ 1 + 1.52 n_u & \text{for } \bar{\lambda}_u \geq 1.3 \end{cases} \quad \text{for ferritic stainless steel,} \quad (12)$$

and

$$k_{vu} = \begin{cases} 1 + 0.25 \bar{\lambda}_v n_v & \text{for } \bar{\lambda}_v < 0.8 \\ 1 + 0.2 n_v & \text{for } \bar{\lambda}_v \geq 0.8, \end{cases} \quad \text{but } k_{vu} \leq 0.6 + \bar{\lambda}_v \quad \text{for } \bar{\lambda}_v < 0.4 \quad (13)$$

in which $\bar{\lambda}_v$ is replaced by $\bar{\lambda}_{TF}$ when torsional-flexural buckling is critical and n_u and n_v are given as follows:

$$n_u = \frac{N_{Ed}}{\chi_u N_{Rk} / \gamma_{M1}} \quad (14)$$

$$n_v = \frac{N_{Ed}}{\chi_v N_{Rk} / \gamma_{M1}}, \quad (15)$$

where χ_v is replaced by χ_{TF} when torsional-flexural buckling is critical. Note that Eqs. (8)–(13) are specified in prEN 1993-1-4 [43] for I-section members based on the research presented in [56,57] as well as mono-symmetric sections that are symmetric about their minor axis and, in the absence of further guidance, are assumed herein to be also applicable to angles. Comparisons of the test and FE ultimate capacities R_u with the resistances predicted by EC3 $R_{b,EC3}$ are made in Table 8 and Fig. 24. The EC3 resistance predictions can be seen to be scattered. The scatter of the results is more significant for low values of θ , and there are a number of predictions on the unsafe side. The conservative predicted resistances for low θ values, i.e. where axial compression is dominant, is owing to the double-counting of the same buckling mode (i.e. torsional and local buckling) in EC3 [2–5]. On the other hand, the lack of consideration of torsional-major-axis-flexural and minor-axis flexural interactive buckling, described in Section 5, results in some predictions on the unsafe side. The EC3 treatment for the simultaneous occurrence of local and lateral-torsional buckling, which becomes dominant for high θ values, has also been shown to be conservative [6,7]. Additionally, in developing the interaction curves for mono-symmetric sections, only cross-section types that are symmetric about the minor axis, like mono-symmetric I-sections (see Fig. 25(a)), were considered [58], which led to the proposal of replacing χ_v with χ_{TF} . In such cross-section types, owing to the position of the shear centre relative to the principal axes [59], the torsional-flexural buckling mode is a combination of torsional and minor-axis flexural buckling, and therefore it is valid to include χ_{TF} in Eq. (9), where in-plane minor-axis bending is dominant. However, for cross-section types where the symmetry is about the major axis, like equal-leg angles (see Fig. 25(b)), the torsional-flexural buckling mode involves torsional and major-axis flexural buckling, hence it would be appropriate to replace χ_u with χ_{TF} in Eq. (8), where in-plane major-axis bending is dominant. Finally, for asymmetric sections under compression, like unequal angles (see Fig. 25(c)), χ_{TF} may replace both χ_u and χ_v since the critical mode is a combination of torsional, major-axis flexural buckling and minor-axis flexural buckling. It is recommended that these changes are also

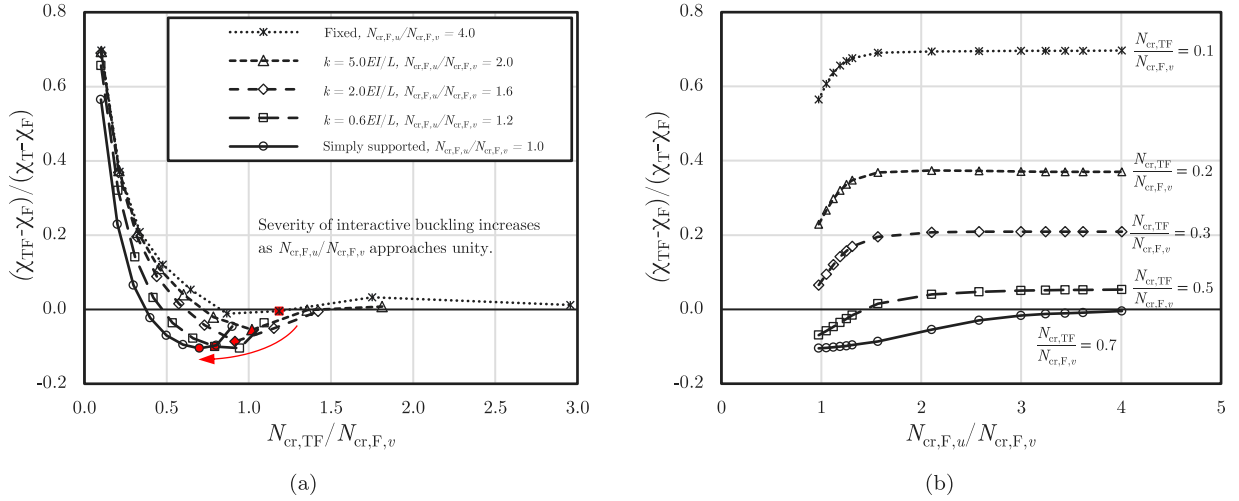


Fig. 23. Variation of $(\chi_{TF} - \chi_F) / (\chi_T - \chi_F)$ with (a) $N_{cr,TF} / N_{cr,F,v}$ and (b) $N_{cr,F,u} / N_{cr,F,v}$ for angle section columns with $\bar{\lambda}_{TF} = 2.5$, where both ends are fixed about the minor axis and elastically restrained about the major axis.

Table 8
Summary of comparisons of test and FE results with EC3 resistance predictions for stainless steel equal-leg angle section columns and beam-columns cylindrically pinned about the major axis.

Source	Manufacturing process	Stainless steel family	Evaluation parameter	$R_u / R_{b,EC3}$
FE	Hot-rolled	Austenitic	Mean	1.33
			CoV	0.23
		Duplex	Mean	1.53
			CoV	0.33
		Ferritic	Mean	1.38
			CoV	0.26
	Cold-formed	Austenitic	Mean	1.57
			CoV	0.21
		Duplex	Mean	1.84
			CoV	0.29
Test	Hot-rolled	Austenitic	Mean	1.70
			CoV	0.24
			Mean	1.22
			CoV	0.04

made to the beam-column design provisions set out in EN 1993-1-1 [60]. New interaction curves, anchored to the end points determined on the basis of the new proposals in Section 7, are needed to provide more accurate resistance predictions for cylindrically-pinned stainless steel equal-leg angle section members under combined compression and major-axis bending.

7. New design proposals

The shortcomings of the current Eurocode 3 design provisions for stainless steel angles were described in Section 6. The lack of an inclusive design method for beam-columns, which is applicable to all cross-section types, along with the absence of interaction curves for equal-leg angles, necessitates the development of new design rules. Recent proposals for the design of stainless steel equal-leg angle section fixed-ended columns [5] and simply-supported beams [7], along with the findings presented in Section 5, are adopted in the current paper to develop new proposals for the design of cylindrically-pinned stainless steel angle section columns and beam-columns with bending about the major axis. The proposed new design formulae are given by:

$$\frac{N_{Ed}}{N_{b,TF,Rd}} + k_{uu} \frac{M_{u,Ed} + \Delta M_{u,Ed}}{M_{b,u,Rd}} \leq 1.0, \quad (16)$$

$$\frac{N_{Ed}}{N_{b,v,Rd}} + k_{vu} \frac{M_{u,Ed} + \Delta M_{u,Ed}}{M_{b,u,Rd}} \leq 1.0, \quad (17)$$

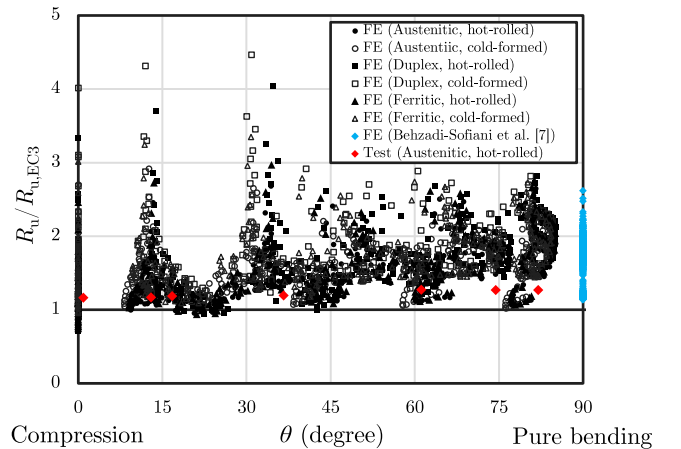


Fig. 24. Comparisons of test and FE results with EC3 ultimate capacities for stainless steel equal-leg angle section columns and beam-columns cylindrically pinned about the major axis.

where $N_{b,TF,Rd}$ and $N_{b,v,Rd}$ are the member buckling resistances under axial compression when torsional-flexural and minor-axis flexural

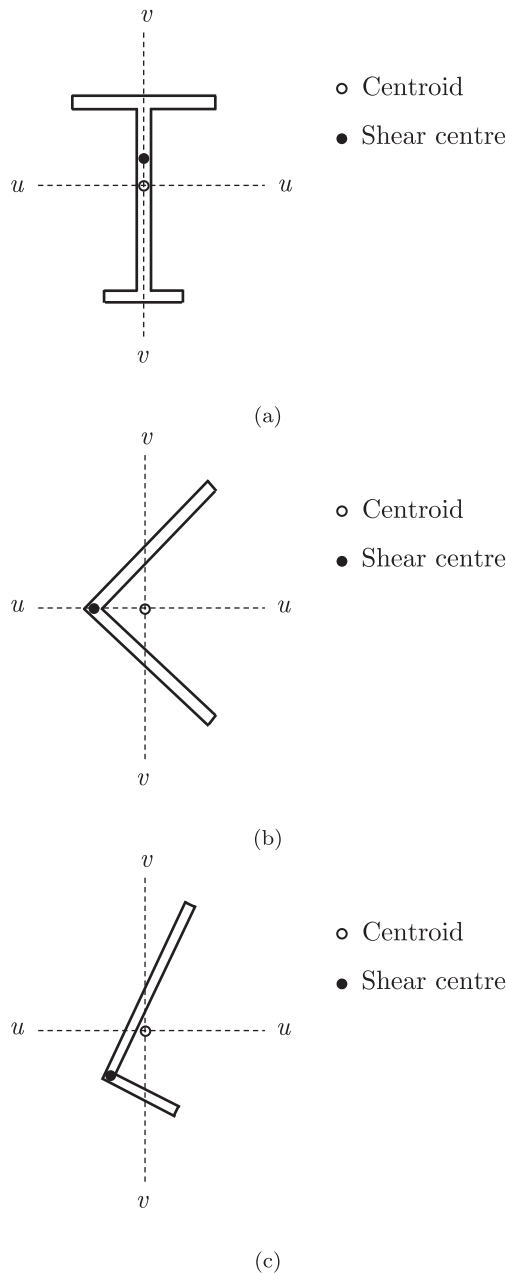


Fig. 25. Illustration of the position of the centroid and shear centre in (a) mono-symmetric I-section, (b) equal-leg angle section and (c) unequal-leg angle section.

buckling is critical, respectively, and $M_{b,u,Rd}$ is the member buckling resistance under bending about the major $u-u$ axis. The new proposals for $N_{b,TF,Rd}$ and $N_{b,v,Rd}$ [5] and $M_{b,u,Rd}$ [7], with some modifications to reflect the findings reported in Section 5, are summarised below.

7.1. Member buckling resistance under axial compression

7.1.1. Torsional-flexural buckling critical (i.e. $N_{cr,TF} \leq N_{cr,F,v}$)

The proposed design expression for determining the buckling resistance of stainless steel equal-leg angle section members under compression [5] when torsional-flexural buckling is critical, is given thus:

$$N_{b,TF,Rd} = \frac{\chi_{TF} A f_y}{\gamma_{M1}}, \quad (18)$$

noting that the gross area A is used for all classes of cross-section [4]. In Eq. (18):

$$\chi_{TF} = \chi_F + \Delta_F (\chi_T - \chi_F), \quad (19)$$

in which the torsional buckling reduction factor χ_T is given by:

$$\chi_T = \frac{\bar{\lambda}_{TF} - 0.188}{\bar{\lambda}_{TF}^2} \quad \text{but} \quad \chi_T \leq 1.0, \quad (20)$$

the flexural buckling reduction factor χ_F is given by:

$$\chi_F = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}_{TF}^2}} \quad \text{but} \quad \chi_F \leq 1.0, \quad (21)$$

and Δ_F is given thus:

$$\Delta_F = \left(1 - \frac{N_{cr,TF}}{N_{cr,F,v}}\right)^p, \quad (22)$$

where

$$p = \begin{cases} 2.0 \bar{\lambda}_{TF} & \text{for } \bar{\lambda}_{TF} \leq 2.0 \\ 2.93 \bar{\lambda}_{TF}^{0.45} & \text{for } \bar{\lambda}_{TF} > 2.0, \end{cases} \quad (23)$$

with the torsional-flexural slenderness $\bar{\lambda}_{TF}$ and ϕ being given thus:

$$\bar{\lambda}_{TF} = \sqrt{\frac{A f_y}{N_{cr,TF}}}, \quad (24)$$

$$\phi = 0.5 \left[1 + \alpha \beta_{\max} (\bar{\lambda}_{TF} - 0.2)^{\beta_{\max}} + \bar{\lambda}_{TF}^2\right], \quad (25)$$

where the parameter $\beta_{\max}$, which accounts for interactive buckling, as described in the section below, is given by:

$$\beta_{\max} = 1.77 - 0.08 \left(\frac{N_{cr,F,u}}{N_{cr,F,v}}\right) \quad \text{but} \quad \beta_{\max} \geq 1.45. \quad (26)$$

For the imperfection factor α , values of 0.6 and 0.49 are recommended for hot-rolled and cold-formed stainless steel angles, respectively. The value of 0.6 is also used for laser-welded sections in prEN 1993-1-4 [43,48], but could be relaxed to 0.76 if retention of the traditional buckling curves (i.e. curve d in this case) was desirable.

7.1.2. Minor-axis flexural buckling critical (i.e. $N_{cr,F,v} < N_{cr,TF}$)

The proposed design expression for determining the buckling resistance of stainless steel equal-leg angle section members under axial compression [5], when minor-axis flexural buckling is critical is:

$$N_{b,v,Rd} = \frac{\chi_F A f_y}{\gamma_{M1}}, \quad (27)$$

where

$$\chi_F = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}_v^2}} \quad \text{but} \quad \chi_F \leq 1.0, \quad (28)$$

in which the normalised slenderness $\bar{\lambda}_v$ and ϕ are given by:

$$\bar{\lambda}_v = \sqrt{\frac{A f_y}{N_{cr,F,v}}}, \quad (29)$$

$$\phi = 0.5 \left[1 + \alpha \beta (\bar{\lambda}_v - 0.2)^{\beta} + \bar{\lambda}_v^2\right], \quad (30)$$

with β being a factor allowing for the influence of interactive (torsional-flexural and minor-axis flexural) buckling. As described in Section 5, the severity of interactive buckling increases as $N_{cr,F,u}/N_{cr,F,v}$ approaches unity; therefore, a higher interaction factor β compared to that adopted for fixed-ended conditions (i.e. when $N_{cr,F,u}/N_{cr,F,v} = 4.0$) in [5] is required for angle section columns cylindrically pinned about the major axis (i.e. when $N_{cr,F,u}/N_{cr,F,v} = 1.0$). The maximum value of the interaction factor, $\beta_{\max} = 1.69$ is adopted for $N_{cr,F,u}/N_{cr,F,v} = 1.0$. A linear relationship between $\beta_{\max}$ and the $N_{cr,F,u}/N_{cr,F,v}$ ratio is proposed for $1.0 < N_{cr,F,u}/N_{cr,F,v} < 4.0$ to reflect the variation in the severity of interactive buckling for different boundary conditions, including

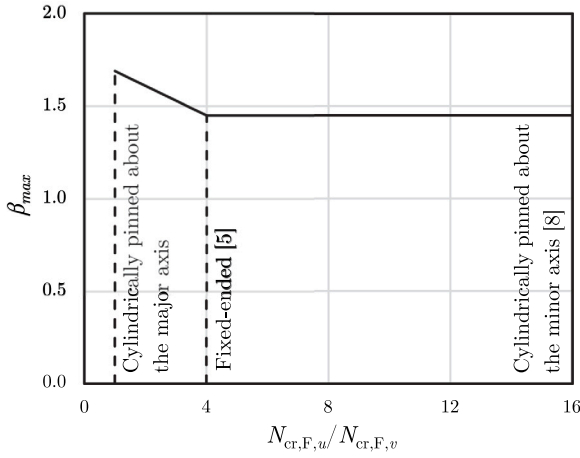


Fig. 26. Relationship between the maximum interaction factor $\beta_{\max}$ and $N_{\text{cr},F,u}/N_{\text{cr},F,v}$.

partially-fixed end conditions. For $N_{\text{cr},F,u}/N_{\text{cr},F,v} \geq 4.0$, $\beta_{\max} = 1.45$. Therefore $\beta_{\max}$ is defined thus:

$$\beta_{\max} = 1.77 - 0.08 \left(\frac{N_{\text{cr},F,u}}{N_{\text{cr},F,v}} \right) \quad \text{but} \quad \beta_{\max} \geq 1.45. \quad (31)$$

The relationship between $\beta_{\max}$ and $N_{\text{cr},F,u}/N_{\text{cr},F,v}$ is illustrated in Fig. 26, with $N_{\text{cr},F,u}/N_{\text{cr},F,v}$ values corresponding to the boundary conditions studied in [5,8] as well as in the current paper being highlighted, noting that the $N_{\text{cr},F,u}/N_{\text{cr},F,v}$ ratio can only vary between 1 and 16 for equal angle section members with different boundary conditions.

The interaction factor β varies with both $N_{\text{cr},TF}/N_{\text{cr},F,v}$ and $N_{\text{cr},F,u}/N_{\text{cr},F,v}$, as described in Section 5. A linear relationship between the interaction factor β and the $N_{\text{cr},TF}/N_{\text{cr},F,v}$ ratio, similar to that proposed in [4], is adopted, but modified to incorporate the variation of β with $N_{\text{cr},F,u}/N_{\text{cr},F,v}$ through $\beta_{\max}$, as given by:

$$\beta = \beta_{\max} + (1 - \beta_{\max}) \left(\frac{N_{\text{cr},TF}}{N_{\text{cr},F,v}} \right) \quad \text{but} \quad 1.0 \leq \beta \leq \beta_{\max}. \quad (32)$$

The relationship between β and $N_{\text{cr},TF}/N_{\text{cr},F,v}$ is shown in Fig. 27 for $N_{\text{cr},F,u}/N_{\text{cr},F,v} = 1.0$ and $N_{\text{cr},F,u}/N_{\text{cr},F,v} \geq 4.0$. Reflecting more severe interactive buckling at lower $N_{\text{cr},TF}/N_{\text{cr},F,v}$ values for cylindrically-pinned angle section columns (for which $N_{\text{cr},F,u}/N_{\text{cr},F,v} = 1.0$) than fixed-ended conditions (for which $N_{\text{cr},F,u}/N_{\text{cr},F,v} = 4.0$), the corresponding β curve (see Fig. 27) has a higher upper bound (i.e. higher $\beta_{\max}$) and reaches the lower bound at a lower $N_{\text{cr},TF}/N_{\text{cr},F,v}$ ratio (i.e. at $N_{\text{cr},TF}/N_{\text{cr},F,v} = 0.7$ for $N_{\text{cr},F,u}/N_{\text{cr},F,v} = 1.0$ compared to $N_{\text{cr},TF}/N_{\text{cr},F,v} = 1.0$ for $N_{\text{cr},F,u}/N_{\text{cr},F,v} = 4.0$).

The imperfection factor remains as specified above: $\alpha = 0.6$ and $\alpha = 0.49$ for hot-rolled and cold-formed stainless steel angles, respectively.

7.2. Member buckling resistance under major-axis bending

For determining the member buckling resistance of stainless steel equal-leg angle section members under major-axis bending, the following expression is proposed [7]:

$$M_{b,u,Rd} = \frac{\chi W_{pl,u} f_y}{\gamma_{M1}}, \quad (33)$$

in which the plastic section modulus $W_{pl,u}$ is used for all classes of cross-section. In Eq. (33):

$$\chi = \chi_{LT} + 4 (\chi_1 - \chi_{LT}), \quad (34)$$

where the local buckling reduction factor χ_1 is given by:

$$\chi_1 = \frac{\bar{\lambda}_{\max,u} - 0.188}{\bar{\lambda}_{\max,u}^2} \quad \text{but} \quad \chi_1 \leq 1.0, \quad (35)$$

the lateral-torsional buckling reduction factor χ_{LT} is given by:

$$\chi_{LT} = \frac{1}{\phi_{LT} + \sqrt{\phi_{LT}^2 - \bar{\lambda}_{\max,u}^2}} \quad \text{but} \quad \chi_{LT} \leq 1.0, \quad (36)$$

and Δ is given thus:

$$\Delta = \begin{cases} \left(1 - \frac{M_{\text{cr},L,u}}{M_{\text{cr},LT,u}} \right)^{3.5} & \text{for } \frac{M_{\text{cr},L,u}}{M_{\text{cr},LT,u}} \leq 1 \\ 0 & \text{for } \frac{M_{\text{cr},L,u}}{M_{\text{cr},LT,u}} > 1, \end{cases} \quad (37)$$

with the maximum normalised slenderness $\bar{\lambda}_{\max,u}$ and ϕ_{LT} being given thus:

$$\bar{\lambda}_{\max,u} = \sqrt{\frac{W_{pl,u} f_y}{M_{\text{cr}}}}, \quad (38)$$

$$\phi_{LT} = 0.5 \left[1 + \alpha_{LT} (\bar{\lambda}_{\max,u} - \bar{\lambda}_{LT,0}) + \bar{\lambda}_{\max,u}^2 \right], \quad (39)$$

where M_{cr} is the minimum of the local $M_{\text{cr},L,u}$ and the major-axis lateral-torsional $M_{\text{cr},LT,u}$ critical buckling moments. The proposed values for the imperfection factor α_{LT} and limiting slenderness $\bar{\lambda}_{LT,0}$ are 0.49 and 0.4, respectively, for both hot-rolled and cold-formed stainless steel angles.

7.3. Interaction factor k_{uu}

The suitability of the existing interaction factors k_{uu} and k_{vu} for use with the new proposed end points of the interaction curve (i.e. the axial and bending resistances set out in Sections 7.1 and 7.2) is assessed herein. The expression for the interaction factor k takes the general bilinear form given thus:

$$k = 1 + (D_1 \bar{\lambda}_i + D_2) n_i, \quad \text{where } i = \{TF, v\}, \quad (40)$$

in which D_1 and D_2 are constants. Eq. (40) can be rearranged into the following expression:

$$\frac{k-1}{n_i} = D_1 \bar{\lambda}_i + D_2 \quad (41)$$

The $(k-1)/n_i$ values corresponding to the test and FE results are plotted against the critical normalised slenderness, $\bar{\lambda}_{TF}$ or $\bar{\lambda}_v$, in Fig. 28. The bilinear EC3 expression for the interaction factor for I-sections is also shown, and can be seen to lie generally on the safe side of (i.e. above) the experimental and numerical results, but excessively so. New expressions were therefore sought to provide a closer match to the data [61]; the following expressions are proposed:

$$k_{uu} = \begin{cases} 1 + (3\bar{\lambda}_{TF} - 1) n_u & \text{for } \bar{\lambda}_{TF} \leq 0.5 \\ 1 + 0.5 n_u & \text{for } \bar{\lambda}_{TF} > 0.5 \end{cases} \quad \text{for austenitic stainless steel,} \quad (42)$$

$$k_{uu} = \begin{cases} 1 + (\bar{\lambda}_{TF} - 1) n_u & \text{for } \bar{\lambda}_{TF} \leq 1.5 \\ 1 + 0.5 n_u & \text{for } \bar{\lambda}_{TF} > 1.5 \end{cases} \quad \text{for duplex stainless steel,} \quad (43)$$

$$k_{uu} = \begin{cases} 1 + (2\bar{\lambda}_{TF} - 1) n_u & \text{for } \bar{\lambda}_{TF} \leq 0.75 \\ 1 + 0.5 n_u & \text{for } \bar{\lambda}_{TF} > 0.75 \end{cases} \quad \text{for ferritic stainless steel,} \quad (44)$$

and

$$k_{vu} = \begin{cases} 1 - 0.35 \bar{\lambda}_v n_v & \text{for } \bar{\lambda}_v \leq 2.0 \\ 1 - 0.7 n_v & \text{for } \bar{\lambda}_v > 2.0, \end{cases} \quad (45)$$

which, when considering k_{uu} , correspond to $D_1 = 3.0$ and $D_2 = -1$ for $\bar{\lambda}_{TF} \leq 0.5$, and $D_1 = 0$ and $D_2 = 0.5$ for $\bar{\lambda}_{TF} > 0.5$ for austenitic stainless steel, and $D_1 = 1.0$ and $D_2 = -1$ for $\bar{\lambda}_{TF} \leq 1.5$, and $D_1 = 0$ and $D_2 = 0.5$ for $\bar{\lambda}_{TF} > 1.5$ for duplex stainless steel, and $D_1 = 2.0$ and $D_2 = -1$ for $\bar{\lambda}_{TF} \leq 0.75$, and $D_1 = 0$ and $D_2 = 0.5$ for $\bar{\lambda}_{TF} > 0.75$ for ferritic stainless steel, and when considering k_{vu} correspond to $D_1 = -0.35$ and $D_2 = 0$ for $\bar{\lambda}_v \leq 2.0$, and $D_1 = 0$ and $D_2 = -0.7$ for $\bar{\lambda}_v > 2.0$ for all families of stainless steel.

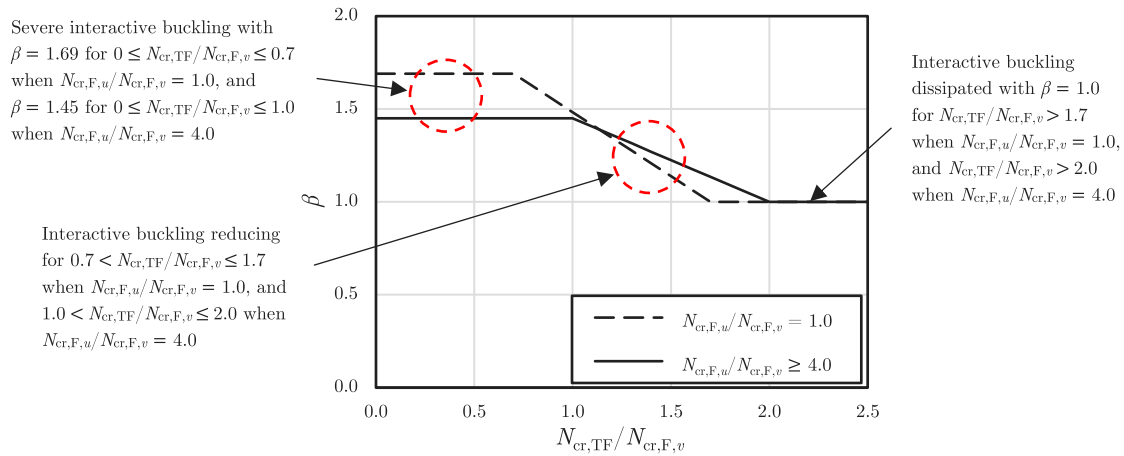


Fig. 27. Relationship between interaction factor β and $N_{cr,TF}/N_{cr,F,v}$ for $N_{cr,F,u}/N_{cr,F,v} = 1.0$ and $N_{cr,F,u}/N_{cr,F,v} \geq 4.0$.

Table 9

Summary of comparisons of test and FE results with resistance predictions according to the new proposals for cylindrically-pinned stainless steel equal-leg angle section members under compression and major-axis bending.

Source	Manufacturing process	Stainless steel family	Evaluation parameter	$R_u/R_{b,prop}$
FE	Hot-rolled	Austenitic	Mean	1.16
			CoV	0.11
		Duplex	Mean	1.19
			CoV	0.12
		Ferritic	Mean	1.19
			CoV	0.11
	Cold-formed	Austenitic	Mean	1.13
			CoV	0.09
		Duplex	Mean	1.18
			CoV	0.12
		Ferritic	Mean	1.18
			CoV	0.10
Test	Hot-rolled	Austenitic	Mean	1.25
			CoV	0.09

7.4. Assessment of design proposals

A summary of the comparisons of the test and FE capacities with the resistance predictions determined according to the new design proposals $R_{b,prop}$ is presented in Table 9 and Fig. 29. By comparing Figs. 24 and 29, it can be seen that the resistance predictions are significantly improved for cylindrically-pinned stainless steel angles under compression and combined loading using the new proposals relative to the current Eurocode 3 provisions.

7.5. Reliability analysis

The reliability of both the prEN 1993-1-4 [43] and newly proposed design rules for cylindrically-pinned stainless steel equal-leg angle section members subjected to compression and combined loading is evaluated in accordance with [62]. As provided in Afshan et al. [63], the material over-strength factor $f_{y,m}/f_{y,n}$ (i.e. the ratio of the mean to the nominal yield strength) and the corresponding CoV of the yield strength V_{fy} were taken as 1.3 and 0.06 for austenitic, 1.1 and 0.03 for duplex, and 1.2 and 0.045 for ferritic stainless steel, while the CoV of the Young's modulus was taken as 0.03. The CoV of the cross-sectional area V_A was obtained based on the parameters of the geometric dimensions provided in Annex E of [60] and adopting the procedure outlined in [63], which led to a V_A value equal to 0.03. Following Annex E of [60], the mean values of the Young's modulus and all the geometric properties were taken to be equal to the nominal values. A summary of the reliability analysis procedure is provided

in [4]. The key reliability analysis results are presented in Tables 10 and 11 for the current EC3 rules and the new design proposals, respectively, where the design fractile factor $k_{d,n}$, the CoV of the test and FE resistances relative to the predictions from the resistance model V_{δ} , and the combined CoV, incorporating the variability of the resistance model and the basic variables V_r , were calculated in accordance with [62]; γ_{M1}^* is the required value of the partial safety factor taken as the mean value for all test and FE results and determined following the approach in [63], while γ_{M1} is the target partial safety factor value (equal to 1.1) and f_a is the acceptance limit [64], given thus:

$$f_a = 1.03 + 0.75(V_r - 0.04) \quad \text{but} \quad 1.03 \leq f_a \leq 1.15 \quad (46)$$

The ratio $\gamma_{M1}^*/\gamma_{M1}$ should satisfy:

$$\gamma_{M1}^*/\gamma_{M1} \leq f_a. \quad (47)$$

A graphical comparison between the obtained $\gamma_{M1}^*/\gamma_{M1}$ values and the acceptance limit f_a is presented in Fig. 30 for the new design method. The current EC3 design expressions clearly lead to unacceptably high γ_{M1}^* values, owing principally to the excessively high scatter of the resistance predictions. On the other hand, the proposed design method leads to $\gamma_{M1}^*/\gamma_{M1}$ ratios less than or near 1.0 for both hot-rolled and cold-formed stainless steel members and is therefore considered to satisfy the Eurocode reliability requirements.

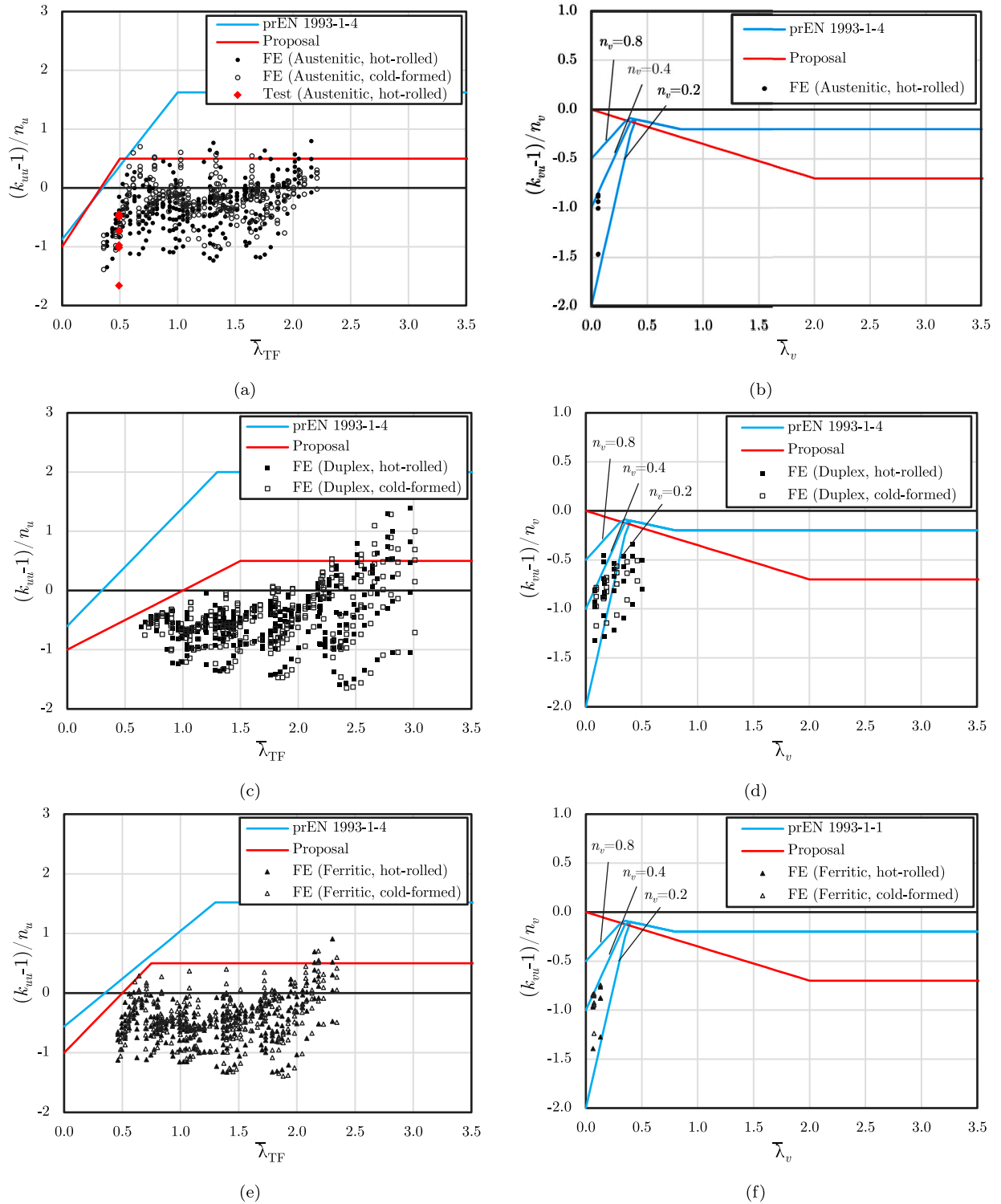


Fig. 28. Comparisons of test and FE results with the existing EC3 and proposed interaction factors k_{uu} and k_{vu} for (a, b) austenitic, (c, d) duplex and (e, f) ferritic stainless steel equal-leg angle section members under combined compression and bending about the major axis.

8. Conclusions

A comprehensive study into the behaviour and design of cylindrically-pinned stainless steel equal-leg angle section members subjected to compression and combined compression and major-axis bending has been presented herein. A programme of physical experiments on hot-rolled austenitic stainless steel angle section columns and beam-columns cylindrically pinned about the major axis and fixed

about the minor axis, together with supporting material tests and initial geometric imperfection measurements, was first reported. Finite element models were then developed and validated against the reported test results; good agreement between the test and FE results was found. A parametric study was subsequently conducted on both hot-rolled and cold-formed angle section columns and beam-columns in austenitic, duplex and ferritic stainless steel covering a spectrum of cross-section

Table 10

Reliability analysis results for cylindrically-pinned stainless steel angle section members under compression and major-axis bending for the current EC3 methodology.

Source	Manufacturing process	Stainless steel family	$k_{d,n}$	b	V_{δ}	V_r	γ_{M1}^*	$\frac{\gamma_{M1}^*}{\gamma_{M1}}$	f_a	$\frac{\gamma_{M1}^*/\gamma_{M1}}{f_a}$
FE	Hot-rolling	Austenitic	3.118	1.326	0.220	0.228	1.38	1.25	1.15	1.09
		Duplex	3.118	1.529	0.302	0.304	1.68	1.53	1.15	1.33
		Ferritic	3.118	1.381	0.240	0.246	1.47	1.33	1.15	1.16
	Cold-forming	Austenitic	3.118	1.565	0.212	0.220	1.14	1.04	1.15	0.90
		Duplex	3.118	1.837	0.275	0.277	1.28	1.17	1.15	1.01
		Ferritic	3.118	1.698	0.231	0.236	1.16	1.05	1.15	0.91
Test	Hot-rolling	Austenitic	3.118	1.220	0.041	0.070	0.81	0.74	1.05	0.70

Table 11

Reliability analysis results for cylindrically-pinned stainless steel angle section members under compression and major-axis bending for the new design proposals.

Source	Manufacturing process	Stainless steel family	$k_{d,n}$	b	V_{δ}	V_r	γ_{M1}^*	$\frac{\gamma_{M1}^*}{\gamma_{M1}}$	f_a	$\frac{\gamma_{M1}^*/\gamma_{M1}}{f_a}$
FE	Hot-rolling	Austenitic	3.118	1.157	0.1183	0.130	1.12	1.02	1.10	0.93
		Duplex	3.118	1.192	0.120	0.126	1.22	1.11	1.09	1.01
		Ferritic	3.118	1.193	0.109	0.122	1.12	1.02	1.09	0.94
	Cold-forming	Austenitic	3.118	1.126	0.089	0.110	1.08	0.98	1.08	0.91
		Duplex	3.118	1.180	0.119	0.125	1.23	1.12	1.09	1.02
		Ferritic	3.118	1.181	0.103	0.117	1.12	1.02	1.09	0.93
Test	Hot-rolling	Austenitic	3.118	1.249	0.094	0.111	0.91	0.82	1.08	0.76

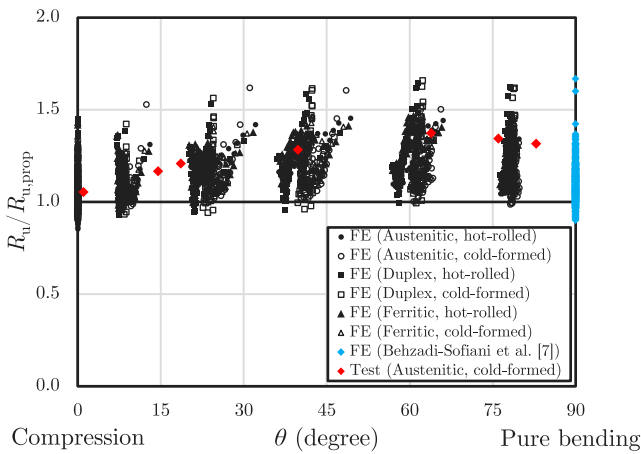


Fig. 29. Comparisons of test and FE results with resistance predictions according to the new proposals for cylindrically-pinned stainless steel equal-leg angle section members under compression and major-axis bending.

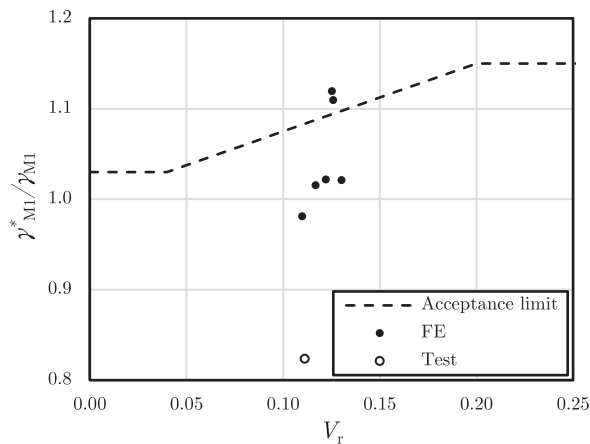


Fig. 30. Comparison of $\gamma_{M1}^*/\gamma_{M1}$ ratios with acceptance limit f_a for new design proposals.

and member geometries and load combinations, with some 2100 numerical results being generated. The FE results, combined with the test results, were used to assess the current EC3 design provisions for stainless steel equal-leg angle section columns and beam-columns cylindrically pinned about the major axis. The resistance predictions were found to be highly scattered relative to the test and numerical data, with predictions both on the unsafe and conservative sides. For column buckling resistance, the current EC3 design provisions account for torsional/local buckling twice, which is the primary source of the observed conservatism. However, for slender columns, the lack of consideration of interaction between torsional-flexural and minor-axis flexural buckling, the severity of which was shown to vary for different boundary conditions and be at the highest level for angles cylindrically pinned about the major axis, leads to unsafe predictions. New expressions were proposed to reflect the variation in the severity of interactive buckling for different boundary conditions through the $N_{cr,F,u}/N_{cr,F,v}$ ratio. For members with mono-symmetric cross-sections, the current EC3 design rules are based on section types that are symmetric about their minor axis. These issues, together with an absence of a specific interaction curve for angle section members under combined compression and major-axis bending, lead to the inaccurate EC3 resistance predictions. A new design approach for cylindrically-pinned stainless steel angle section members under compression and combined loading, addressing the above shortcomings has been proposed. Overall, the new design proposals have been shown to lead to significantly more accurate and reliable resistance predictions for both hot-rolled and cold-formed stainless steel equal-leg angle section columns and beam-columns cylindrically pinned about the major axis.

CRediT authorship contribution statement

Behnam Behzadi-Sofiani: Writing – original draft, Validation, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **M. Ahmer Wadee:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization. **Leroy Gardner:** Writing – review & editing, Validation, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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